# Metal displacement

Student: ..... Class: .....

## Growing metal crystals

**1.** A shiny copper wire was suspended in a test tube containing a solution of silver nitrate. Over time, shiny silvery coloured crystals grew on the surface of the copper wire. The experiment was repeated with a shiny magnesium wire in place of the copper wire. Once again silvery crystals grew on the wire surface. However, the crystals grew at a faster rate.

(a) Identify the silvery crystals that formed in each experiment.

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(b) Write word equations for each reaction.

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(c) Explain why the crystals grew faster in the second experiment.

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## The activity series of metals

**2.** The following table shows the activity series of metals.

| Li | K | Na | Ca | Mg | Al | Mn | Cr | Zn | Fe | Ni | Sn | Pb | H | Cu | Hg | Ag | Au | Pt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

The photo shows zinc reacting with hydrochloric acid.



Photo: Public domain

(a) Identify the gas released in the reaction.

(b) Write a word equation for the reaction.

(c) Explain why this reaction is an example of a displacement reaction.

(d) Use the activity series to identify:

(i) metals that will not react with hydrochloric acid

(ii) the metal that will react most rapidly with the acid solution.

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